

Rahul Kolla

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Summary

Software Intern and AI Developer with early-career experience building autonomous AI agents and production-grade LLM infrastructure. Delivered AI-driven workflow automation for SAP-based platforms, integrating ReAct agents, multi-tool orchestration, and prompt engineering using LangChain, FastAPI, and OpenAI/Claude APIs. Developed scalable backend services and REST APIs that support real-time data operations and enterprise AI pipelines. Seeking to apply agentic AI expertise to create innovative, high-impact AI solutions.

Technical Skills

- **Agentic AI & LLM Systems:** AI Agents, ReAct Agents, Multi-Agent Workflows, Tool Calling, RAG Pipelines, Prompt Engineering, Context Management, LangChain, LlamaIndex, OpenAI APIs, Claude APIs, Ollama, Hugging Face, Vector Databases
- **Programming & Backend:** Python, TypeScript, JavaScript, Java, SQL, FastAPI, Flask, REST APIs, Node.js, PostgreSQL, SQLite, Redis
- **AI/ML:** LLM Applications, NLP, Semantic Search, Embeddings, Deep Learning, CNNs, LSTMs
- **Infrastructure & DevOps:** Docker, AWS, Railway, GitHub Actions, CI/CD, Linux, Git/GitHub, Ansible, Webhooks, Structured Logging, Secrets Management

Experience

AiFA Labs – Cerebro SASA Platform

Sep 2025 - Present

Software Intern

Charlotte, NC

- Developed workflow automation systems using Python, Docker, and LLM-powered reasoning to integrate backend APIs and orchestrate SAP-based pipelines, enabling faster task completion and reducing manual effort
- Built and refined AI workflows using LangChain, TypeScript, and webhooks to orchestrate tools, engineer prompts, process structured data, and execute tasks autonomously within an Agile team, which streamlined development cycles and lowered manual intervention

AiFA Labs

Jan 2024 - Jul 2024

Backend Developer Intern

Hyderabad, India

- Built scalable backend services and REST APIs using Python and Flask for AI-driven enterprise workflows and real-time data operations, enabling faster data processing and reducing system latency
- Automated ingestion and logging pipelines, improving operational efficiency and supporting production-ready AI platform features.

Projects

Helmsman – Autonomous DevOps AI Agent

- Built an autonomous DevOps agent using ReAct-style reasoning loops and Claude tool-use APIs to monitor repository health, dependency risks, CI failures, and PR activity. Integrated GitHub Actions, OSV.dev CVE scanning, and Slack APIs into orchestrated multi-tool workflows with automated issue generation, reporting pipelines, token budgeting, context management, and tool-result caching.

FERRET – Autonomous AI Research Agent

- Developed an agentic AI system that autonomously performs web research, synthesizes multi-source information, and generates structured reports with live progress tracking, tool orchestration, and export pipelines.

AI-Powered Smart Document Intelligence System

- Engineered a local RAG platform using FastAPI, LangChain, Ollama, and SQLite for semantic document retrieval, summarization, and offline LLM-powered querying.

Automated Security Pipeline for SIEM and IDS

- Designed a real-time security monitoring pipeline using Ansible, Wazuh, and Suricata to automate distributed threat detection and infrastructure observability.

AI-Based Tumor Heterogeneity Detection (Patented)

- Proposed and implemented an AI-driven approach for quantifying intra-tumor heterogeneity using image-based biomarkers for healthcare diagnostics.

Education

University of North Carolina at Charlotte | *Master of Science, Computer Science*

Charlotte, NC 2024 - 2026

Vellore Institute of Technology – AP | *Bachelor of Technology, Computer Science*

Amaravati, India 2020 - 2024

Certifications

- Hugging Face Agents Course: Hugging Face, 2025
- Deep Learning Institute: NVIDIA, 2024
- Microsoft Azure Fundamentals (AZ-900): Microsoft, 2024
- Prompt to Prototype: Google, 2025